"Scalability challenges at the edge can be addressed by creating independent failure domains and Kubernetes' declarative infrastructure"

Keith Basil,
General Manager (Edge Business Unit), SUSE

It's been a decade since Google released Kubernetes. Almost every single cloud provider has since embraced this open source container orchestration system. In recent years, there has been a shift from using cloud and Kubernetes concepts to pushing that API to the edge. In an exclusive conversation with OSFY's ***Yashasvini Razdan, Keith Basil, General Manager (Edge Business Unit), SUSE,*** elucidates how organisations can fulfil the stringent demands of flexibility and scalability in edge computing through Kubernetes.

Q. Why do we need containers for the edge?

A. Containers became essential at the edge, not only for their lightweight and standardised deployment but also for solving the management-at-scale challenge that edge computing introduces. Kubernetes, which originated in the cloud, standardised infrastructure management for microservices and containerised applications. In 2019, Rancher released K3s, a lightweight version of Kubernetes, designed to run on low-resource hardware with a single command, offering a CNCF-certified API. K3s quickly gained traction among hobbyists and developers using Kubernetes on devices like the Raspberry Pi and Intel NUC. PoCs emerged for running Kubernetes on small, remote machines, attracting attention from industries like retail and manufacturing.

As edge use cases grew, the focus shifted from managing a few Kubernetes clusters to supporting thousands. Containers, with Kubernetes' declarative nature, allowed teams to reuse cloud-based tools, processes like CI/CD pipelines, and developer skills to meet the scaling demands at the edge, while maintaining cloud-like efficiency.

Q. How is deploying Kubernetes at the edge different from deploying it on the cloud?

A. In the cloud, the infrastructure is preconfigured, so developers only interact with the API, without needing to worry about hardware or infrastructure setup. In contrast, at the edge, you need to build up